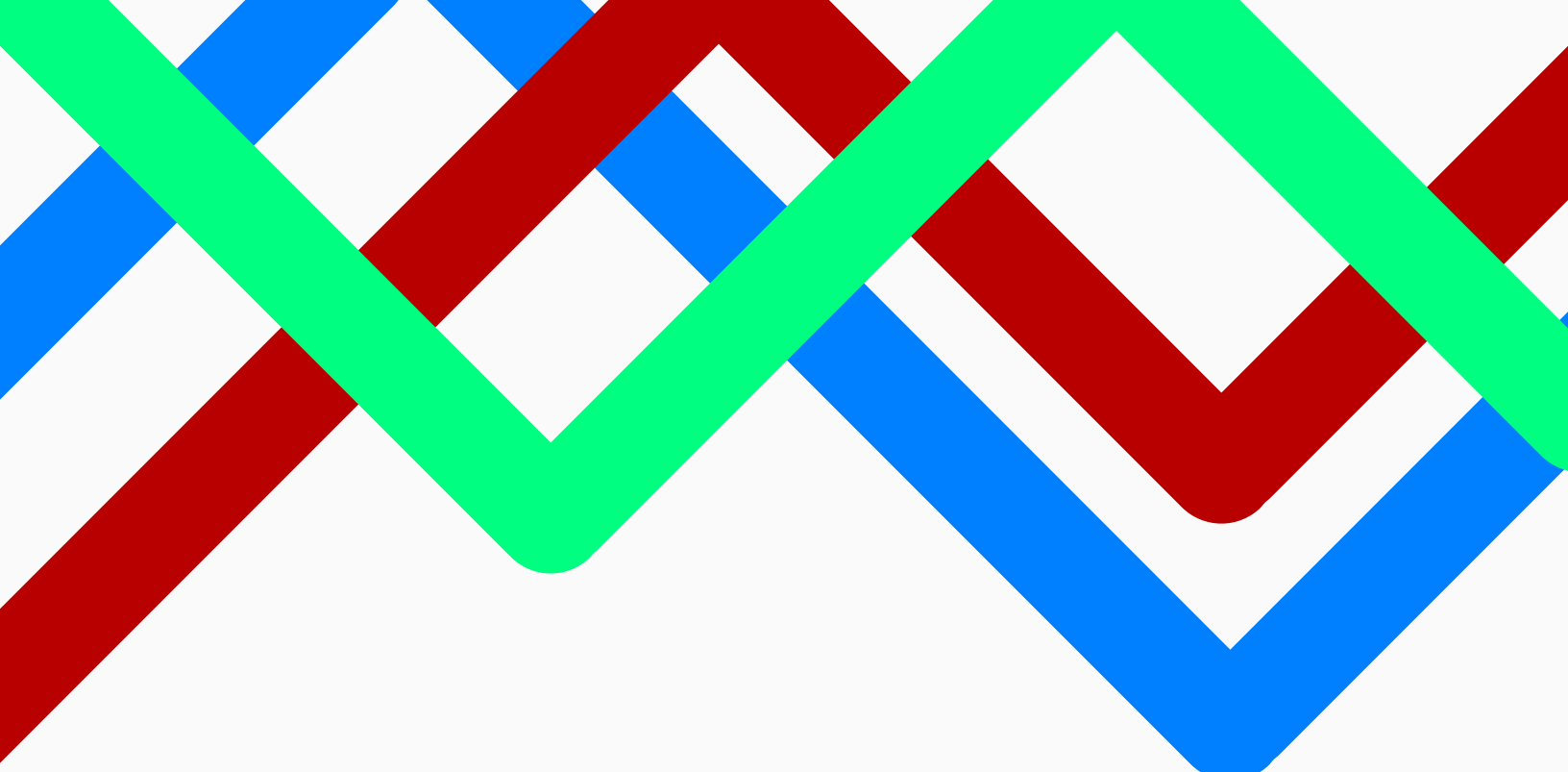




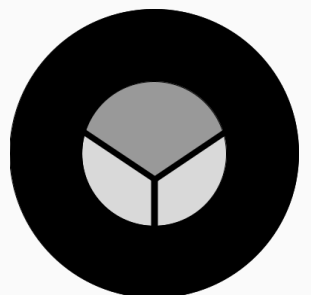
Cracker Protein Lab

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DE Intro to Biology I

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“Truth is ever to be found in the simplicity, and not in the multiplicity and confusion of things” - Issac Newton



1 Abstract

To show the denaturing of amylase through the change in pH levels and how the efficacy and efficiency of enzyme performance are scale-based.

2 Background / Introduction

3 Question / Problem

How does ***amylase***, an enzyme found in saliva, break polysaccharides into simpler carbohydrates in different PH levels?

4 Elements & Hypothesis

4.1 Hypothesis

If the pH levels of three solutions of pH {4, 5.5, 7}, are exposed to the enzyme *amylase* found in saliva and mixed with a polysaccharide containing cracker, then there should be a trend whereas the pH 7 solution will have the highest density of glucose, the pH 5.5 solution should have the medium density of glucose, and the pH 4 solution should have the lowest density of glucose.

4.2 Experimental Elements

- Independent Variable: PH Level of the solution
- Dependent Variable: Glucose concentration produced (in mmol/L)
- Experimental Groups:
 - Group 1: Saltine Cracker, Saliva, pH 4.0 solution
 - Group 2: Saltine Cracker, Saliva, pH 5.5 solution
- Positive Control Group:
 - Saltine Cracker, Saliva, optimal pH solution (pH 7.0)
- Negative Control Group:
 - Saltine Cracker, optimal pH solution (pH 7.0)
- Controls
 - Saliva to Solution ratio
 - Cracker to Solution ratio
 - Time allowed for enzymatic processes
 - Method of glucose measurement

5 Materials / Set up

5.1 Materials

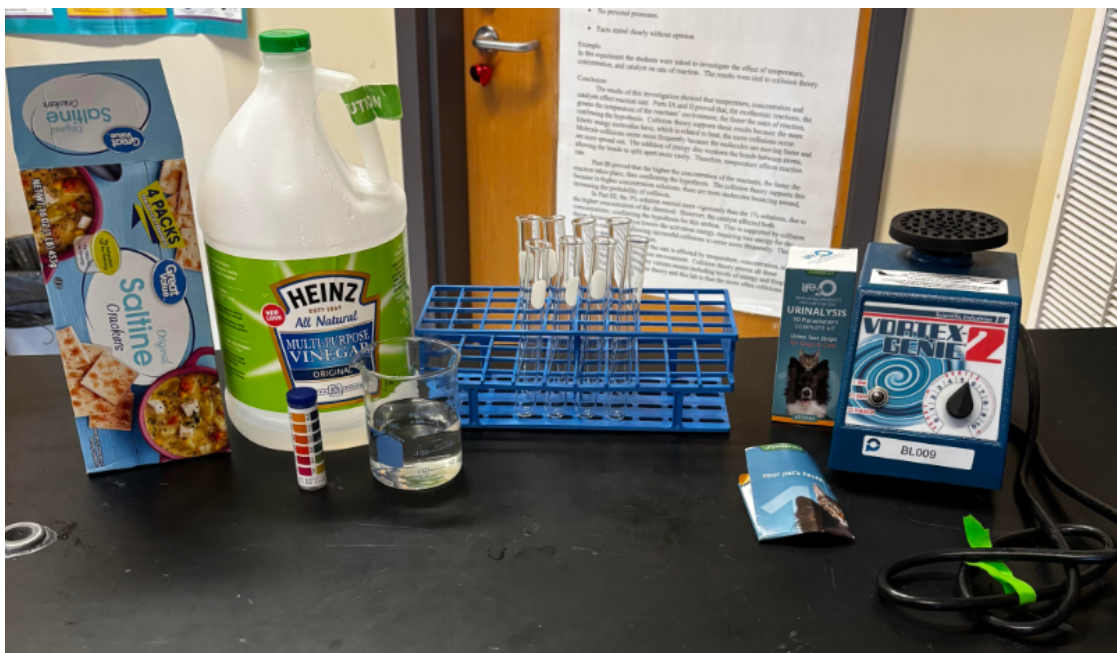
The following materials were gathered:

1. 4 saltine cracker pieces (approximately 1 gram each)
2. Roughly 100 mL of Heinz all-natural vinegar
3. pH test strips
4. Roughly 250 mL of distilled water
5. 8 15-25mL test tubes
6. Test tube rack
7. Life2O Pet Urinalysis Test Strips: Dog and Cat Urine Test Strips
8. Genie 2 Vortex Mixer SI-0236
9. Fresh Human Saliva (Roughly 20 mL total)
10. Stopwatch
11. Mortar & Pestle
12. 4 stirring rods
13. Plastic gloves
14. Tape
15. Black Marker
16. 2x 250 mL beaker
17. Plastic test tubes

5.2 Set Up Procedure

1. All lab materials were collected and thoroughly inspected to ensure cleanliness.
2. Two sets of four test tubes were labeled with {"C","pH=7","pH=5.5","pH=4"} using tape and a marker.
3. The first set of four test tubes were filled with 10 mL of distilled water.
 - The water was filled up in the first beaker with about 100 mL of distilled water.
 - The vinegar was filled up to about the 50 mL mark of the second beaker.
4. Then, according to the test tube labels, the first set of four had 2 mL of vinegar added incrementally (until the pH was correct) and checked via a pH strip.
5. When all four test tubes of the first set were at the pH level indicated by each respective label, the tubes were put away, in a row, back onto the test tube rack
6. All lab members put on their gloves.

An exemplar image of an example setup without pH solutions can be found below:



6 Methods / Procedure

1. Saliva was collected by each individual labmate into their respective plastic test tube. This was to minimize the safety impact of having different contaminants on each test tube.
2. Four crackers (approximately 1 gram each) were crushed by a mortar and pestle into a fine powder.
3. The four crackers were distributed evenly across the **second** set of test tubes. (So now, the second set only has cracker powder in it.)
4. The saliva was mixed into a fourth plastic test tube and vortexed via the Genie 2 Vortex Mixer
5. The labmates were careful here; the saliva was then distributed evenly into the **first** set of test tubes. (So now, the first set only has solution + saliva. Only $\frac{3}{4}$ contained saliva because of the negative control.)
6. Carefully, the first set of test tubes was added onto each respective test tube of the second set and began a stopwatch.
7. This is where the rush began. During the 30 seconds of curing (determined via the stopwatch), each test tube was vortexed to ensure that each solution was at least somewhat homogenous. This was a step where caution was taken to ensure that mixture escaped the test tube.
8. After the 30 seconds of curing (determined via the stopwatch) was over, one urinalysis strip was placed into each test tube to allow for glucose measurement.
9. The stopwatch was reset and began counting to 30 seconds again. After the 30 seconds were over, each urinalysis strip was carefully removed from each test tube. Then, the urinalysis strip was compared to the package, and the glucose levels were measured and recorded.

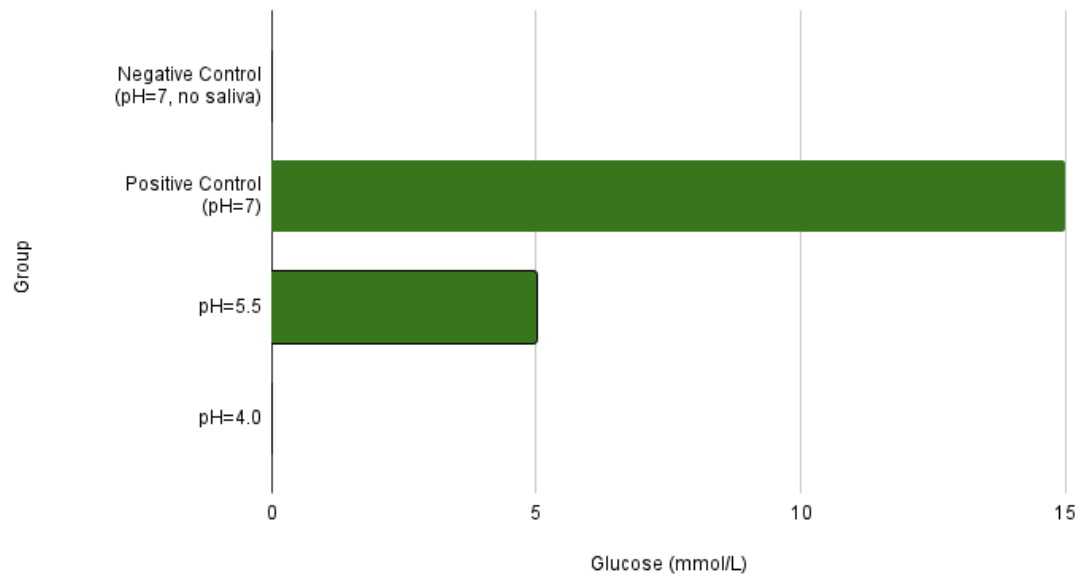
7 Data / Results

7.1 Table

Group	Glucose Measured (mmol/L)
Negative Control (pH=7.0, NO saliva)	0
Positive Control (pH=7.0)	15
pH=5.5	5
pH=4.0	0

7.2 Graph

Glucose (mmol/L) versus Group



8 Conclusion / Discussion